

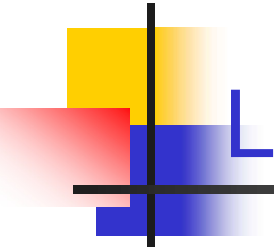
**Université Abdel-Hamid Mehri
Constantine II**

Faculté NTIC/DIFA 3° TI

Installation et Administration of Network Services

Linux Commands

R. NOUARA

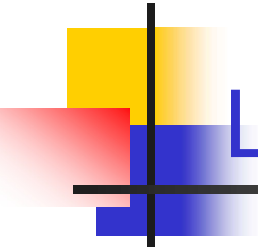


Linux Commands

- **uname** : displays information about the system's kernel, including the kernel name, hostname, kernel release, kernel version, and machine hardware name.

`uname -a`

- **whoami** : whoami command in Linux returns the current user's username.



Linux Commands

- **man** : display the manual of any command

`man cmd`

- **whatis** : command displays a short description of a command or program.

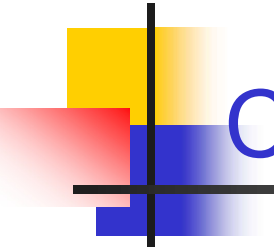
`whatis cmd`

- **pwd** : prints the full name (the full path) of current/working directory.

- **clear** : clear our terminal (**Ctrl+I**).

- **history** : Display the list of executed commands

- **hostname** : Display the hostname of the system (the file `/etc/hostname`)



Commands for files management

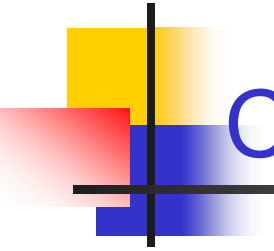
- **cat** : display the content of a file on screen.

`cat /etc/hosts.conf`

- **less** : similar to cat but display per page,

- **touch** : create an empty file.

`touch /home/monficher.txt`



Commands for files management

- **ls** : lists the contents of a directory

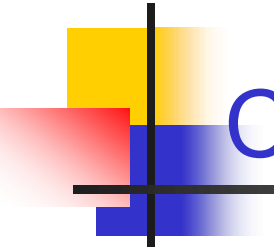
ls [options] [files]

- **We can use generic characters:**

- ***** :replace a string of characters.
- **?** : replace one character

- **Options:**

- **-a** Show all (including hidden)
- **-R** Recursive list
- **-r** Reverse order
- **-t** Sort by last modified
- **-S** Sort by file size
- **-l** Long listing format
- **-1** One file per line



Commands for files management

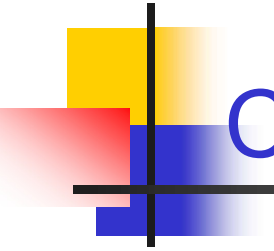
- **cp** : copies file(s) or directories to a destination directory.

cp [irv] source destination

cp /etc/*.conf /var

- **Option:**

- -i :interactive.
- -r :recursive.
- -v :verbose.



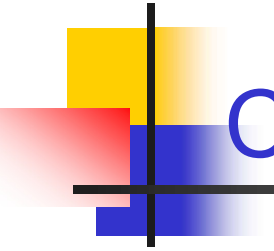
Commands for files management

- **mv** : move file(s) to destination directory.

cp [ifv] source destination

- **Option:**

- -i :interactive.
- -f :force.
- -v :verbose.



Commands for files management

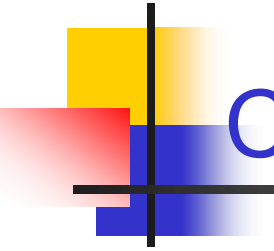
- **rm** : remove file(s)

`rm [ifvr] fichiers`

- **Option:**

- `-l` :interactive.
- `-f` :force.
- `-v` :verbose.
- `r` : récursive.

`rm file1 ..... fileN`



Commands for files management

- **mkdir** : create a directory

`mkdir [p] dir`

- **Option:**

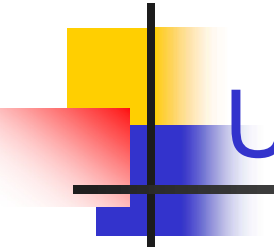
- `-p` : Create parent directories if they dont exixsts.

`mkdir -p directory1/sub1/sub2`

`directory1`

`|____sub1`

`|____sub2`



User management

- **useradd (adduser) : Adding a user**

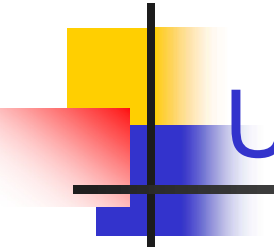
**useradd <username> [-g Grp] [-G
Grp1[,grp2,...,Grpn]]**

useradd malik -g mongroup

- Create a user and assign him a default directory in /home

- **getent passwd**

Give users list



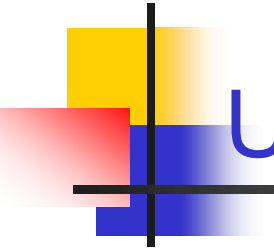
Users Commands

- **passwd** : activate a user count by assigning a password.

passwd <username>

The command add a line in the file /etc/passwd

[username]:[x]:[UID]:[GID]:[Comment]:[Home_D]:[Default shell]



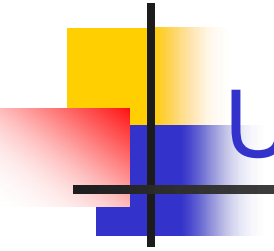
Users Commands

- **groupadd** : Adding a group

groupadd <group-name>

- Create a group of users.
- The command add a line in **/etc/group**

[Grp _name]:[Grp_password]:[GID]:[Grp_members]

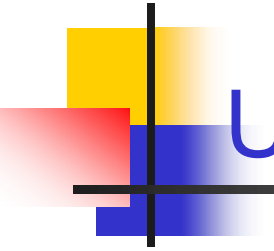


Users Commands

- **usermod** : modify a user.

usermod [options] <username>

- **-g** add user to group .
- **-d Rep_pers**: change home directory.
- **-e DATE_END_VALIDITY**
- **-G GRP -G GR1[,GR2,...[,GRN]]**
- **-L** : lock the password



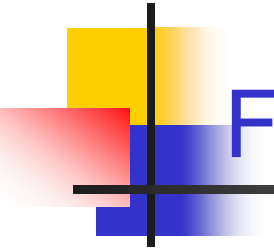
Users Commands

- **userdel** : delete a user and its associates files

`userdel <username>`

- **groupdel** : delete a group

`groupdel <username>`

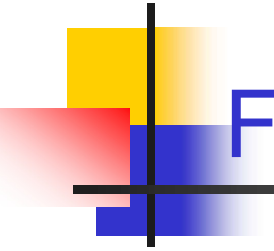


Files Access

- Permission: composed by three bits rwx.
- **r**: read
- **w**: write
- **x**: execution
- If the bit is set **1** the permission is active, a **0** not activate.
- Can be written in decimal:
- **5** correspond to :

Read and execution

r	w	x
1	0	1

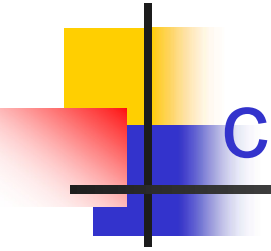


Files Access

- Example:

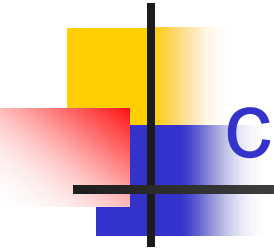
rw-rw-r-- 1 user users 150 Mar 19 08:08 f1.doc

- **rw-** : for user **user** (read+write = 6)
- **rw-**: for users group (**users**) (read+write = 6).
- **r--**: for others. (read = 4).
- The permission of f1.doc is **664**



chmod

- The permissions are attributed to three user's categories **ugo** (**u***ser*, **g***roup*, **o***ther*).
- **u** : user
- **g**: The group(s) which the user belongs,
- **o** : other users.



chmod

- Change permissions of file(s).

chmod [option] new_mod files

- **New_mod** : 3 digits that specify the privileges of ugo.
- Example :

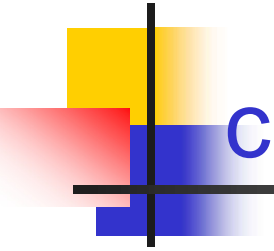
■ **chmod 764 /home/rep.sh**

rwX

rw-

r--

- **Option R** : recursive.



chown

- Change the owner and/or the group of file(s).

chown [option] new_owner files

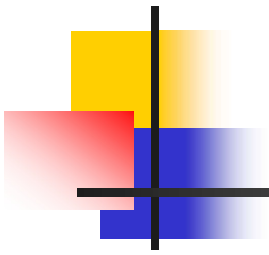
- **new_owner**: the new owner of the file.

chown user yum.conf

- Change the owner of yum.conf to **user**

chown user:clients yum.conf

- Change the owner and the group of yum.conf.



THANK YOU